From the figure, it is clear that number of white tiles in each row are 1, 2, 3, 4, …, 12 which clearly forms an Arithmetic Progression.

Similarly the number of blue tiles in each row are 0, 1, 2, 3, …, 11 which is also an Arithmetic Progression.

$$\text{Number of white tiles } = 1+2+3+\cdots+12 = \frac{12}{2}[1+12] \ = 78$$

$$\text{Number of blue tiles } = 0+1+2+3+\cdots+11 = \frac{12}{2}[0+11] \ = 66$$

$$\text{The total number of tiles in the mosaic } = \ 78 + \ 66 \ = 144$$

Example 2.38 The houses of a street are numbered from 1 to 49. Senthil's house is numbered such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number?

Solution Let Senthil's house number be x.

It is given that $1+2+3+\cdots+(x-1) = (x+1)+(x+2)+\cdots+49$

$$1+2+3+\cdots+(x-1) = \left[1+2+3+\cdots+49\right]-\left[1+2+3+\cdots+x\right]$$

$$\frac{x-1}{2}\left[1+(x-1)\right] = \frac{49}{2}\left[1+49\right]-\frac{x}{2}\left[1+x\right]$$

$$\frac{x(x-1)}{2} = \frac{49\times 50}{2}-\frac{x(x+1)}{2}$$

$$x^2-x = 2450-x^2-x \Rightarrow 2x^2 = 2450$$

$$x^2 = 1225 \Rightarrow x = 35$$

Therefore, Senthil's house number is 35.

Example 2.39 The sum of first n, $2n$ and $3n$ terms of an A.P. are S_1, S_2 and S_3 respectively. Prove that $S_3 = 3(S_2 - S_1)$.

Solution If S_1, S_2 and S_3 are sum of first n, $2n$ and $3n$ terms of an A.P. respectively then

$$S_1 = \frac{n}{2}\left[2a+(n-1)d\right],\ S_2 = \frac{2n}{2}\left[2a+(2n-1)d\right],\ S_3 = \frac{3n}{2}\left[2a+(3n-1)d\right]$$

Consider, $S_2 - S_1 = \frac{2n}{2}\left[2a+(2n-1)d\right]-\frac{n}{2}\left[2a+(n-1)d\right]$

$$= \frac{n}{2}\left[\left[4a+2(2n-1)d\right]-\left[2a+(n-1)d\right]\right]$$

$$S_2 - S_1 = \frac{n}{2}\times\left[2a+(3n-1)d\right]$$

$$3(S_2-S_1) = \frac{3n}{2}\left[2a+(3n-1)d\right]$$

$$3(S_2-S_1) = S_3$$

Thinking Corner

1. What is the sum of first n odd natural numbers?
2. What is the sum of first n even natural numbers?

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